

ABIR DAS

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ABOUT ME

AI/ML Engineer with strong research and practical experience in computer vision, self-supervised learning, NLP, and medical imaging. First author on 4 AI research papers (2 published, 2 ongoing), skilled in building deep learning models using PyTorch and TensorFlow. Experienced in deploying scalable AI solutions with Hugging Face, Flask, and AWS. Passionate about explainable AI and advancing real-world applications in healthcare and speech processing.

WORK EXPERIENCE

NINANO Company Inc

(Sep 2024 - Nov 2024)

Head. 2F, 315-11, Hyeoksin-ro, Gimcheon-si, Gyeongsangbuk-do, Republic of Korea.

AI Software Developer Intern

Accomplishments:

- **Developed an Advanced Stereo Vision System:** Built a stereo vision solution using MIDAS and SGBM deep learning algorithms from scratch, enabling obstacle detection within 6 meters and ensuring safe drone operation at a consistent height of 10 meters.
- **Integrated Object Detection and Safety Mechanisms:** Implemented YOLO for real-time object detection and integrated it with the drone's flight controller, enabling autonomous safety braking and operator notifications.
- **Developed Autonomous Recovery System:** Designed an algorithm for transitioning the drone between brake and loiter modes, ensuring seamless mission continuity.
- **Key Technologies:** NVIDIA Jetson Orin Nano, YOLO, Python, PX4 Autopilot, OpenCV, Raspberry Pi

Sub-Project:

Enhanced Gimbal Camera Functionality: Modified YOLO- based object detection algorithms for a drone-mounted gimbal camera to autonomously track a designated target while simultaneously identifying other objects within the frame.

Key Technologies: YOLO (You Only Look Once), Python, Computer Vision, Gimbal Control Systems.

SpaceK

(Dec 2022 - Jan 2023)

Chungnam National University Start-up Support Center, 160 Techno 2-ro, Yuseong-gu,

Daejeon, Republic of Korea.

Embedded Systems Intern

Accomplishments:

- Designed and implemented embedded software solutions for satellite communication systems, improving data transmission efficiency by 20% and enhancing control reliability.
- Developed microcontroller-based solutions using Embedded Linux, UART, I2C, and SPI protocols to streamline satellite communication processes.

RESEARCH EXPERIENCE

Research Assistant | MSPI Research Group | [01-2023] – [10-2024]

- Conducted research in computer vision and deep learning, focusing on real-world AI applications.
- Worked on projects related to Machine Fault Diagnosis, applying AI techniques like YOLO, MiDaS, or Explainable AI.
- Assisted in writing and reviewing research papers, including one published publication.

Site - <https://sites.google.com/view/drjiauddin/research-areas?authuser=0>

Research Assistant | Chungnam National University | [12-2024] – [Present]

- Contributed to research on brain-computer interface (BCI) using motor imagery EEG data and Funds
- Co-authored the paper “Advanced Machine Learning Models for Motor Imagery Analysis in Brain-Computer Interfaces” with the professor as a corresponding author
- Worked on model design, signal preprocessing, and manuscript development

PROJECTS

MediScan – Skin Disease Detection via Mobile Camera

Built a mobile-friendly AI system to detect 7+ skin diseases using a CNN classifier. Integrated Grad-CAM for explainability. Deployed on Hugging Face with real-time UI. Tech: PyTorch, OpenCV, Streamlit, Grad-CAM, Hugging Face

Real-Time Accent Change Generator (Ongoing)

Developing a system to convert Indian-accented English speech to US-accented voice in real time using Tacotron2, Whisper, and audio synthesis. Tech: PyTorch, Whisper, WaveGlow, Tacotron2, Flask, Streamlit

GitHub Resume Website

Designed and deployed a modern personal portfolio using GitHub Pages. Features projects, publications, and contact links. Tech: HTML, CSS, GitHub Pages

Real-Time Object Detection on 52-Card Deck

Developed a real-time machine learning model to detect and classify standard playing cards using OpenCV and contour analysis. Accuracy: 98% · Tech: Python, OpenCV

JCFS Assistant Chatbot

Created a departmental assistant chatbot using Flask and Google NLP API for automated query handling. Integrated sentiment analysis to prioritize critical student concerns.

Impact: Reduced manual workload by 50% · Tech: Flask, Google NLP, Sentiment Analysis

Car Parking Occupancy Detection

Built a computer vision system to monitor parking slot availability using YOLO and real-time camera input. Accuracy: 90% · Tech: YOLOv3, OpenCV, Python

Boston Housing Data Visualizer

Designed an interactive dashboard to analyze and visualize Boston housing prices using AI-driven clustering and regression algorithms. Tool: Tableau · ML: K-Means, Linear Regression

Facial Expression Recognition

Built a CNN model to classify human emotions into 7 categories using FER-2013 dataset.

Accuracy: 92% · Tech: TensorFlow, Python

Scalable ML Model Deployment on AWS

Trained and deployed a scalable ML pipeline using AWS SageMaker, reducing training time by 40% and enabling remote inference. Tech: AWS SageMaker, Python, scikit-learn

SeoulSpotlight – Tourism Website

Launched a mobile-friendly, SEO-optimized WIX website to showcase Seoul’s cultural landmarks.

Tool: WIX

EDUCATION

Woosong University, South Korea

March 2021 - February 2025

Bachelor of Science in Artificial Intelligence - 3.83 - 4.5 / 86.6%

Siliguri Government Polytechnic, India

August 2016 - July 2019

Diploma in Electronics & Instrumentation Engineering - 70%

Kanchrapara Harnett High School

January 2011 - March 2016

High School - 79%

SKILLS & EXPERTISE

Languages

Python, SQL, Bash

Frameworks & Libraries

PyTorch, TensorFlow, Keras, scikit-learn, OpenCV, Transformers (Hugging Face), CLIP, Grad-CAM, CUDA

Machine Learning & AI Techniques

CNNs, RNNs, Vision Transformers (ViT), Transfer Learning, Self-Supervised Learning, Few-Shot & Zero-Shot Learning, Explainable AI (XAI), Contrastive Learning, Object Detection (YOLO, MiDaS, SGBM), Sentiment Analysis

Tools & Platforms

Google Colab, Jupyter, Anaconda, Git, VS Code, Tableau, SPSS, GitHub

Deployment & APIs

Flask, AWS SageMaker, Hugging Face Spaces, GitHub Pages, REST APIs, Streamlit

Hardware & Embedded Systems

NVIDIA Jetson Orin Nano, Raspberry Pi 5, PX4 Autopilot, Embedded Linux

CERTIFICATIONS

- **Introduction to TensorFlow for AI, ML, and DL** - Coursera
 - **Neural Networks and Deep Learning** - Coursera
 - **Machine Learning with Python (IBM)** - Coursera
 - **Digital Accounting Transformation** – Partnership of Asian Management School
 - **Python** – Partnership of Asian Management School
 - **Python Programming** – Microsoft, Kolkata (Aug 2019)
 - **Android App Development** – IIT Kharagpur (Nov 2019)
 - **Data Visualization** – Partnership of Asian Management School
 - **UX Design** – Partnership of Asian Management School
 - **Elice Game Development in Python** – Korea
 - **Research Assistant Certificate** – Chungnam National Univ. Prof JJ Kim
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PUBLICATIONS

- **Machine Fault Diagnosis Using Sensor Data and Explainable AI Techniques** – Computers, Materials & Continua (CMC)Journal - <https://www.techscience.com/cmc/v80n3/57907>
 - **Enhanced EEG Signal Classification in Brain Computer Interfaces Using Hybrid Deep Learning Models** – Scientific Reports -
 - **Few-Shot and Zero-Shot Brain Tumour Classification using CLIP & ViT** - Under review at International Journal of Sensors, Wireless Communications and Control
 - **Self-Supervised Multi-Label Vision Transformer for Chest X-Ray Diagnosis** - To be submitted to a journal with IF > 8
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